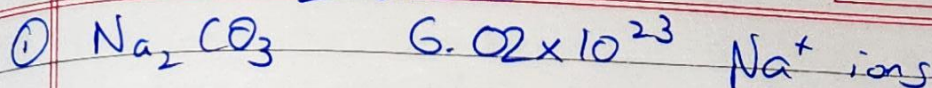


# Test

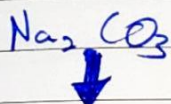
Date |

Page |



$$\text{Moles of Na}^+ = \frac{6.02 \times 10^{23}}{6.02 \times 10^{23}} = 1$$

1 mole of  $\text{Na}_2\text{CO}_3$



$$\text{Moles of Na}_2\text{CO}_3 = \frac{1 \text{ mole of Na}}{2}$$

$$= 0.5 \text{ moles}$$

$$\begin{aligned} \text{Molar Mass} &= 46 + 12 + 48 \\ &= 106 \end{aligned}$$

$$\begin{aligned} \text{Mass} &= \text{moles} \times \text{molar mass} \\ &= 53 \text{g} \end{aligned}$$

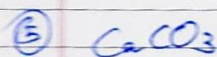
④

②  
③



|    |    |    |
|----|----|----|
| 2  | 2  | 3  |
| x3 | x3 | x3 |
| 6  | 6  | 9  |

⑤



GMA = 100g

$\therefore \text{Ca} = 40$

$\therefore \frac{\text{Ca}}{\text{GMM}} \times 100$

$\frac{40}{100} \times 100 = 40\%$

④

② Because the volume can change if pressure or temperature changes.

⑦ It is the mass of an atom compared to  $\frac{1}{12}$ th part of ~~an~~ C-12 atom

⑤ ~~A mole of an atom is defined as the amount of~~

③  ~~$V = 5.6 \text{ dm}^3$~~   
 $m = 12$   
 $V = 22.4$   
 At STP molar

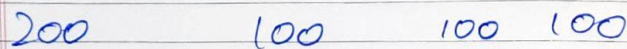
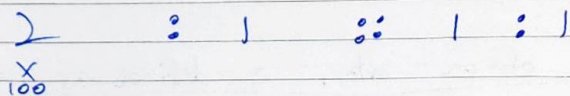
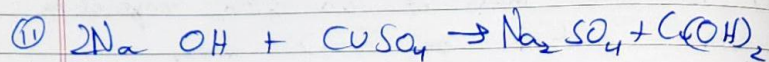
~~RMM~~ =  $\frac{\text{Mass}}{\text{moles}}$

Moles =  $\frac{V}{\text{Molar Mass}}$   
 $= \frac{5.6}{22.4} = 0.25$

$\therefore \text{RMM} = \frac{\text{Mass}}{\text{moles}} = \frac{12}{0.25} = 48$



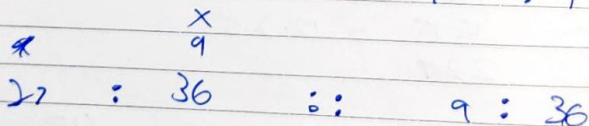
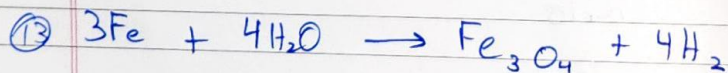
(10) Atomic weights are fractional as they are averages of all the isotopes of the given element.



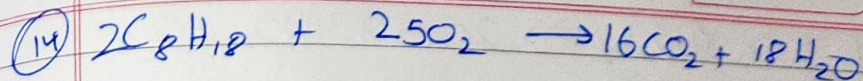
(12) 100g of  $\text{Cu(OH)}_2$

It is pale blue colored

~~13~~



27g of Fe is needed



(i) 8 moles of  $\text{CO}_2$

(ii)  $8 \times 22.4 = 179.2 \text{ dm}^3$

(iii) Mass =  $16 \times 44$   
= 704g

$$\begin{array}{r} 44 \\ \times 16 \\ \hline 64 \\ + 64 \\ \hline 704 \end{array}$$

(2) Mass of  $\text{NH}_3 = 68\text{g}$

~~RAM~~

~~GMM~~ =

GMM = 17g

Molar Volume = 22.4 L

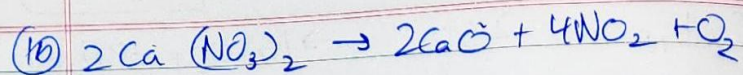
(1)

Moles =  $\frac{\text{Mass}}{\text{GMM}} = \frac{68}{17} = 4$

(ii) Volume =  $4 \times 22.4 = 89.6 \text{ dm}^3$

(iii) Molecules =  $4 \times 6.022 \times 10^{23}$   
=  $24.088 \times 10^{23}$



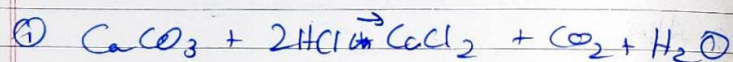


$$\begin{array}{r} 2 \\ \times \\ 41 \\ \hline 82 \end{array} \quad \therefore \quad 2 : 4 : 1$$

$$82 \quad \therefore \quad 82 : 164 : 41$$

$\therefore$  82g of CaO is produced  
and 164 g of

(12) ~~Molar Mass~~



(11) Mass of 4.5 moles of  $\text{CaCO}_3$

$$\begin{aligned} \text{Mass} &= 4.5 \times 100 \\ &= 450\text{g} \end{aligned}$$

(12) Volume =  $\frac{450 \times 22.4}{100}$   
 $= 100.8\text{dm}^3$

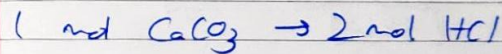
$$\begin{array}{r} 224 \\ \times 45 \\ \hline 1120 \\ + 900 \\ \hline 10080 \end{array}$$

$$\begin{array}{r} 224 \\ \times 45 \\ \hline 1120 \\ + 900 \\ \hline 10080 \end{array}$$

(13) 1 mol of  $\text{CaCO}_3 = 1\text{mol}$  of  $\text{CaCl}_2$

$$\begin{aligned} \text{Mass} &= 4.5 \times 111 \\ &= 499.5\text{g} \end{aligned}$$

(14) Moles of HCl used



$$\therefore \text{Moles of HCl} = 4.5 \times 2 = 9\text{mol}$$

(18)

- (i) B<sub>m</sub>
- (ii) A<sub>m</sub>
- (iii) No limiting agent
- (iv) B<sub>m</sub>
- (v) A<sub>m</sub>

(19) 9

(i) Weight = 4.6g

$\therefore$  No. of atoms

$$\frac{\text{Gram atoms}}{23} = \frac{4.6}{23}$$

$$= 0.2 \text{ gram atoms}$$

(ii)  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

$$= [64 + 32 + 64] + [5 \times (16 + 2)]$$

$$= [128 + 32] + [5 \times 18]$$

$$= 160 + 90$$

$$= 250 \text{g}$$

$$\therefore \frac{90}{250} \times 100 = 36\%$$



Empirical Formula =  $\text{XY}_2$



Empirical Formula weight =  $x + 2y$

Molecular mass =  $2 \times$  Empirical formula (weight)

$$= 2 \times (x + 2y)$$

$$= 2x + 4y$$

$$2(x + 2y)$$

$$= 2(x + 2y)$$

$$= \text{X}_2\text{Y}_4$$